

CMU-SV · 49797 · SPRINT 4 KICKOFF · SEP 14

SSB

Student Second Brain

The open-source alternative to **UniFlow Study**.
Your course material, your lectures, your notes –
indexed, cited, and on your own machine.

TODAY Two decisions I need signed off – or argued down – before the Sprint 4 architecture doc on Sep 22.

01

A semester is scattered across forty decks and a notebook you *stopped updating in week three.*

Course material lives in a syllabus PDF, dozens of slide decks, assignment pages, and announcements. Lectures evaporate the moment they end. And the tools that fix this are proprietary, metered, and keep your recordings on someone else's servers.

THE INCUMBENT UniFlow Study — \$39.20/mo at the top tier. The free tier caps you at 30 AI conversations per month.

02

Three pillars.

01 / ASK

Grounded answers

Every claim cited back to the exact page, file, or moment in a lecture. If your material doesn't support an answer, it says so instead of guessing.

02 / CAPTURE

Lectures, kept

At class time it makes the page, records, transcribes locally with Whisper, and puts a notes editor beside the transcript. Both get indexed.

03 / GENERATE

Study material

Mock tests, mindmaps, slides, flashcards – built from what's actually in your courses, each traceable to its source.

NOTE Pillar 3 is the one UniFlow doesn't have. It's our wedge, and it came out of our own whiteboard session.

03

Where we win, and *where we don't.*

	UNIFLOW STUDY	SSB
License & cost	Proprietary · \$0-\$39.20/mo · metered	Open source · self-hosted · unmetered
Where data lives	Vendor cloud	Your machine
Grounded cited Q&A	Yes — "UniMind"	Yes
LMS sync	Canvas, Blackboard, Moodle	Canvas (CMU-first)
Transcription	Deepgram · hosted · bilingual	Whisper · local
Mock tests · mindmaps · flashcards	No	Yes
Auto session notes pages	Partial	Yes — on your class schedule
Writing assistant	Yes	Not yet
Grades in a vector store	Unknown	Never — read live

HONEST READ They're broader today. We win on ownership, cost, and generating material rather than only answering questions.



From ~~Canvas AI Tutor~~ to *SSB*.

WAS — SPRINTS 1-3

A Socratic tutor over one course

Guides rather than answers. One shared index per course, public to everyone enrolled. Scope: answer questions about course material.

NOW — SPRINTS 4-9

A per-student second brain across all your courses

Ask, capture, generate. One private index per student. Scope: everything you'd otherwise keep in a notebook you abandoned.

- **A Socratic-only tutor is a feature, not a product.** Nobody opens an app to be asked questions when they're behind.
- **Our own whiteboard named the real product.** Transcribe, mock test, mindmaps, notes, flashcards — that was on the board, and none of it was in the spec.
- **Socratic survives as a toggle.** Answer-first is the default. We keep the pedagogy without betting the product on it.

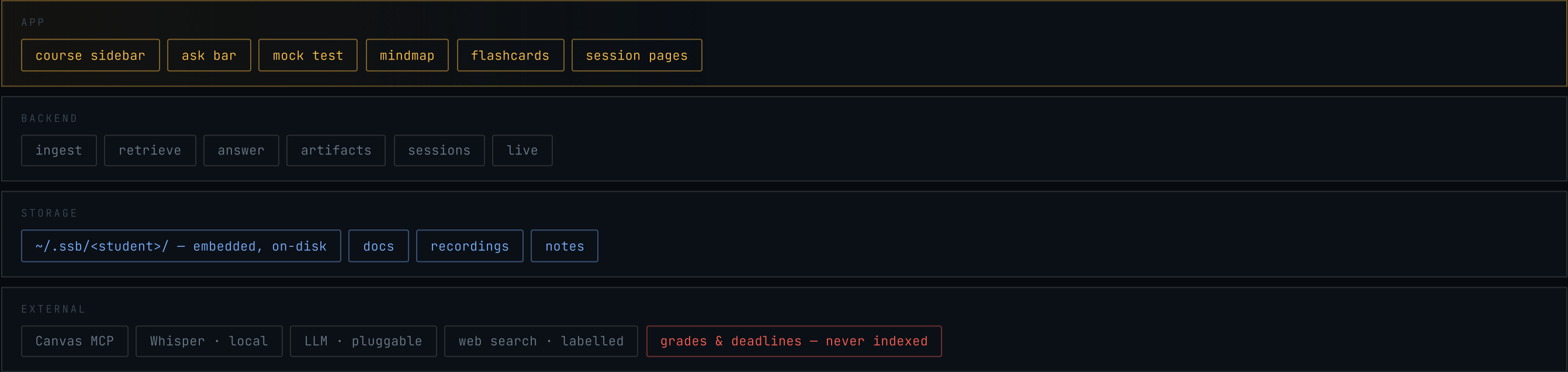
05

We built it on Onyx. *It worked.* We're dropping it anyway.

1,068 LINES · NATIVE CANVASCONNECTOR	v4.7.3 ONYX STANDARD · DEPLOYED	Live PAGES · ASSIGNMENTS · ANNOUNCEMENTS
01 · LICENCEFATAL The permissions are paid Permission sync resolves through <code>fetch_versioned_implementation</code> to a commercially-licensed tier. An open-source product cannot have a permission model that needs a commercial licence. That's a contradiction, not a trade-off.	02 · ARCHITECTURE It's one shared corpus Onyx is built as a single shared index with ACLs on top. We need one private index per student. That's either one Onyx per student — 11 containers each — or no enforcement at all.	03 · DEPLOYMENT Eleven containers OpenSearch, Postgres, Redis, MinIO, two model servers, and more — 10-16 GB of Docker RAM. Our competitor is a desktop app. "Install Docker" is not a student onboarding flow.

SPRINT 3 VERDICT — MODIFY Keep RAG, change the substrate. It took three fixes just to survive a student token, and it still never reached modules or files — where most CMU course content actually lives. A documented negative result is worth the full 20% "Findings and Project Decision" weighting.

Ours, top to bottom.



ONE INDEX PER STUDENT - PHYSICALLY

Not a filter on a shared store. A query-filter bug leaks someone else's lecture recordings; a missing directory just returns nothing. Worth the duplicated embeddings.

PERSONAL DATA IS NEVER INDEXED

Grades, deadlines and submissions are read live from Canvas with your own token, per request. Always current, and never sitting in an embedding store.

07

Nine weeks. *Five components.*

SPRINT CADENCE

S3	Sep 15	Feasibility & baseline ✓
S4	Sep 22	Architecture & implementation plan
S5	Sep 29	Core prototype – ingest, retrieve, cited Q&A
S6	Oct 6	Alpha – session capture, study artifacts
S7	Oct 20	User & impact validation
S8	Oct 27	Robustness & responsible AI
S9	Nov 17	Beta & independent testing
–	Dec 1	Product, impact & defence

COMPONENTS – CLAIM ONE

Ingestion & parsing Canvas pull · PDF/PPTX · chunking · the OCR risk	OPEN
Retrieval & citations embeddings · hybrid search · source tracking	OPEN
App & session pages sidebar · ask bar · notes editor	OPEN
Capture & transcription scheduler · Whisper · transcript indexing	OPEN
Artifact generation mock tests · mindmaps · flashcards · grounding evals	OPEN

Three things.

1 Ratify killing Onyx — or argue it down

I want a verbal yes or no in this room. Silence isn't agreement, and I don't want this reopened in Sprint 6 when it costs us the alpha.

2 Claim a component

Five on the previous slide. Sprint 4's architecture doc is due Sep 22 and it needs names against each one.

3 Pick a licence

"Open source" is our central claim against UniFlow, so this gets settled now rather than left dangling in the README.

MIT

Apache-2.0

AGPL-3.0 — the pointed choice

THEN Objections. That's what the rest of the hour is for.

09